UNIT 5 · TOPIC 10(B)

Visual Range, Instrument Layout and Visual Limitations

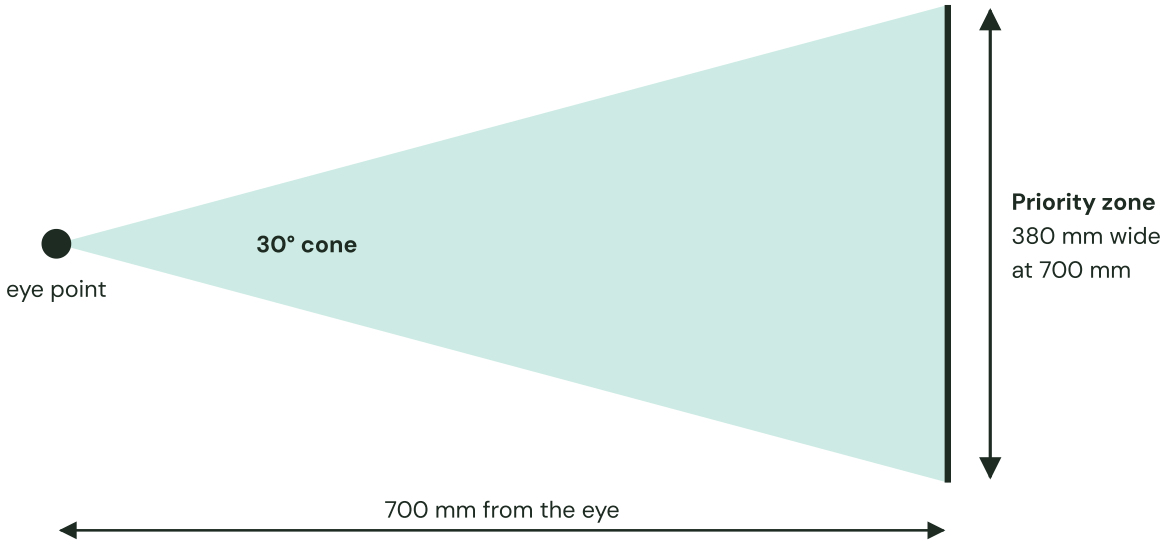
Syllabus topic: Visual range and limitations; seat design

Visibility requirements (p. 226)

Type	Requirement
PRIMARY ("out the window")	Look in ANY direction: NEAR-GROUND vision to front and rear (watching implement and hazards) plus far vision all round
SECONDARY	Instruments and controls inside the workplace

Both visibility and clearance derive from the operator's anthropometric and biomechanical characteristics.

Instrument-panel guidelines (Purcell 1980, p. 229)



Central priority zone — the 30° easy-eye-movement cone (redrawn from the source figure)

1. Group instruments and controls of similar function.
2. Make related groups equal in size.
3. Design the panel as symmetrically as possible.
4. Provide a CENTRAL PRIORITY ZONE — the arc within the **30° cone** of easy eye movement, **380 mm wide at 700 mm** from the operator's eye.
5. Group gauges horizontally by function — **ENGINE gauges LEFT** of panel centre, **TRANSMISSION gauges RIGHT** — with the centre marked by the steering column, tachometer or indicator lights.
6. Keep the remaining instruments in the same relative positions on every vehicle in the line.
7. Standardise panel controls across the line likewise.

- **SAE J209** covers instrument face design and location. Electronic/digital clusters are replacing dials (p. 230).

Visual limitations to remember

- Blind zones behind and beside the tractor — mirrors, cameras, reversing alarms.
- Glare and night lighting.
- Dust and vibration blur instrument reading.
- Ageing operators lose accommodation — use larger characters and higher contrast.
- Colour-blind users — code by shape and position as well as colour.

Practice MCQs

Q133. Instrument face design and location is covered by:

(a) SAE J209 (b) SAE J898
(c) ASAE S310.3 (d) ISO 500

Q134. Highest-priority instruments should lie within the eye-movement cone of:

(a) 10° (b) 30°
(c) 60° (d) 90°

Q135. That priority zone is how wide at 700 mm from the operator's eye?

(a) 150 mm (b) 280 mm
(c) 380 mm (d) 600 mm

Q136. Per Purcell's panel rules, engine gauges are placed:

(a) Right of panel centre (b) Left of panel centre
(c) Behind (d) On the fender

Q137. Transmission gauges are placed:

(a) Left of centre (b) Right of centre
(c) Below the seat (d) Outside

Q138. Primary tractor visibility requires especially:

(a) Near-ground vision to front and rear (b) Sky vision
(c) Under-engine view (d) Rear-wheel view only

Answers

Q133 — (a) SAE J209
Q134 — (b) 30° — *p. 229 — 30° easy-eye-movement cone.*
Q135 — (c) 380 mm — *2 × 700 × tan 15° ≈ 375 ≈ 380 mm.*
Q136 — (b) Left of panel centre
Q137 — (b) Right of centre
Q138 — (a) Near-ground vision to front and rear

Sources: Liljedahl, Turnquist, Smith & Hoki — Tractors and their Power Units, Ch. 9 (spatial/visual/control requirements pp. 226–231; seating pp. 219–224); SAE J899 / J1163 / J898 / J185 / J154a / J209; ASAE S335.3, S304.5 / SAE J389b, EP443; Purcell (1980), Stikeleather (1981); NPTEL lectures.